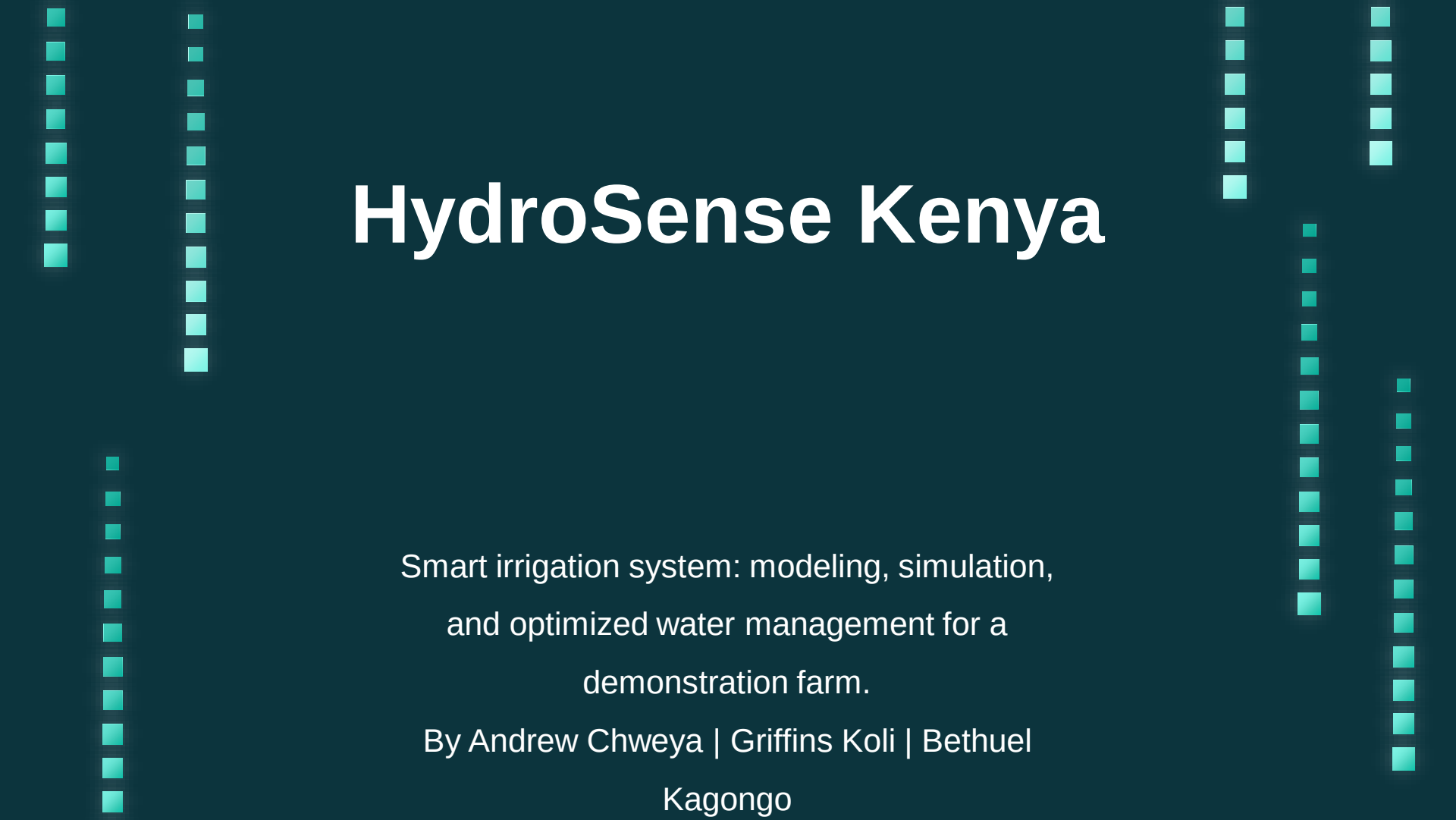




HydroSense Kenya

Smart irrigation system: modeling, simulation,
and optimized water management for a
demonstration farm.

By Andrew Chweya | Griffins Koli | Bethuel

Kagongo

Introduction

- HydroSense-Kenya delivers an end-to-end scientific computing workflow for *smart irrigation*, combining data ingestion, numerical modeling, simulation, and optimization to reduce water use while protecting crops.
- The system demonstrates **12% water savings**, uncertainty-aware decision support, and production-ready numerical methods suitable for deployment in smallholder and research farm contexts.



01

Topic 1



Project Overview

- HydroSense-Kenya frames water scarcity challenges and defines a computational solution using a discrete water-balance model.
- It targets operational decision support for farms, emphasizing *practical deployment*, measurable water reductions, and resilience to rainfall uncertainty.



Data & Inputs

Datasets: daily weather, multi-zone soil sensors, and crop parameters.

Cleaning handled missing values, outliers, and sensor errors; descriptive stats informed model calibration and uncertainty quantification.



Results Summary

- HydroSense-Kenya achieved **12% water savings** versus baseline through optimized scheduling, reducing pump time by 18% and saving ~17,000 L over 30 days.
- Monte Carlo uncertainty shows a 34% shortage risk and 95th-percentile demand of 231,000 L, highlighting rainfall as the dominant risk driver.

Methods & Models

Core model: discrete *water-balance* $S(t+1)=S(t)+R+I-ET-D$.

Evapotranspiration uses a simplified empirical formula. Numerical methods: **RK4**, Euler, root-finding (Newton, bisection, secant), and linear allocation via Gaussian elimination.

Simulation & Optimization

30-day simulations compared Euler vs RK4; RK4 preferred for stability.

Monte Carlo (10,000 runs) quantified shortage/over-irrigation probabilities. Optimization minimized water subject to moisture constraints, favoring frequent, small applications.

Implementation & Validation

Modular codebase with notebooks, src/, and tests; 94% coverage across 65+ tests.

AI-assisted tooling aided test generation and docs. Numerical safeguards use tolerance-based comparisons to prevent floating-point errors.



Conclusions

- HydroSense-Kenya demonstrates a production-ready approach for *uncertainty-aware* irrigation: measurable water savings, robust numerical methods, and validated software practices.
- Recommend RK4 for deployment, real-time rainfall updates for risk reduction, and incremental field trials for scaling.

Thanks!

Do you have any questions?

youremail@freepik.com

+34 654 321 432

yourwebsite.com



CREDITS: This presentation template was created by [Slidesgo](#), and includes icons by [Flaticon](#), and infographics & images by [Freepik](#)